Herrera_Christian_Lab11

April 17, 2026

1 Lab 11.Data Visualization (Questions)

2 Assignment Submission Guidelines

2.0.1 Steps to Submit Your Lab Assignments Using Google Colab

1. **Submission Deadline:**
   - All assignments must be submitted **no later than 23:59 PM tonight**.
   - Late submissions will not be accepted unless prior arrangements have been made by the TAs.
2. **Complete Your Work:**
   - Finish your work in Google Colab and ensure that your notebook is saved.
3. **Submit Your Assignment on Canvas:**
   - Log in to **Canvas** and navigate to the assignment submission page.
4. **File Naming Convention:**
   - Please name your files as follows: `Lastname_Firstname_AssignmentName`
   - Example: `Alex_John_Lab11.ipynb` `Alex_John_lab11.pdf`.
5. **Technical Issues:**
   - If you encounter any technical issues with Canvas or your submission, please contact the TAs immediately **before the deadline** to avoid penalties.

3 Questions

3.1 Please use the churn dataset to answer these questions.

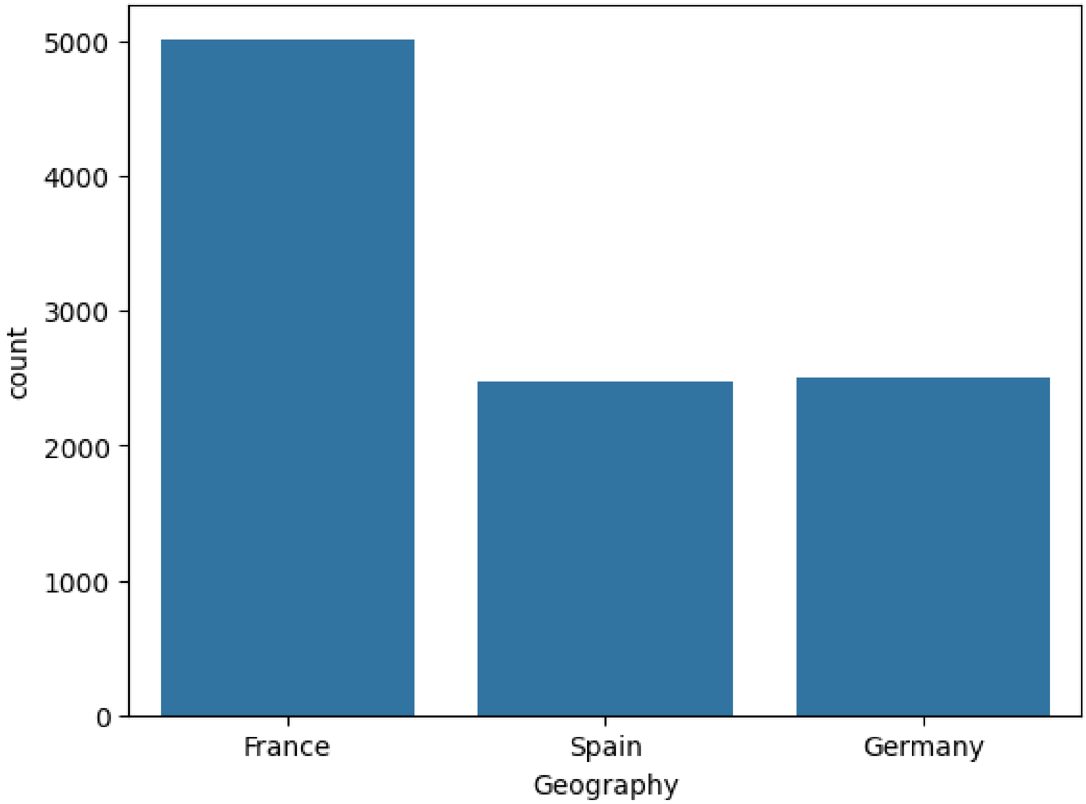
3.1.1 Question 1

3.1.2 Create a bar chart to visualize the distribution of 'Geography' among customers.

[73]:
```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_csv('Churn.csv')
```

```
sns.countplot(data=df, x='Geography')
```

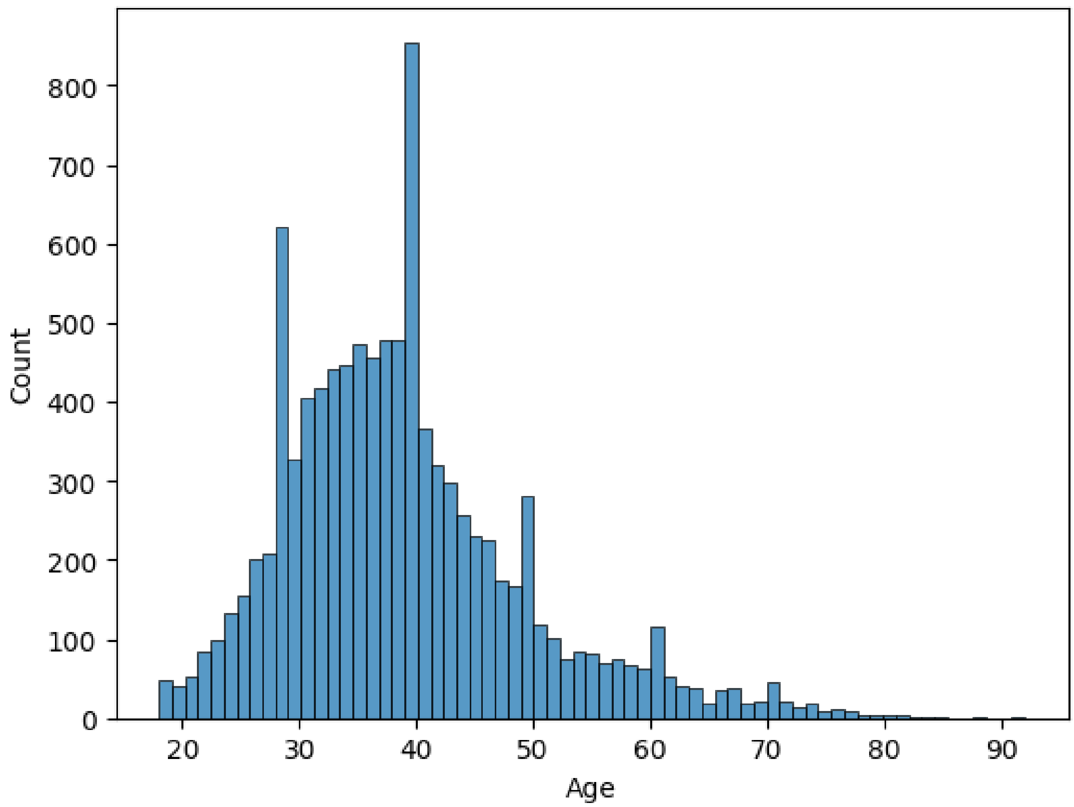
[73]: <Axes: xlabel='Geography', ylabel='count'>



3.1.3 Question 2

3.1.4 Create a histogram to visualize the distribution of 'Age' among customers

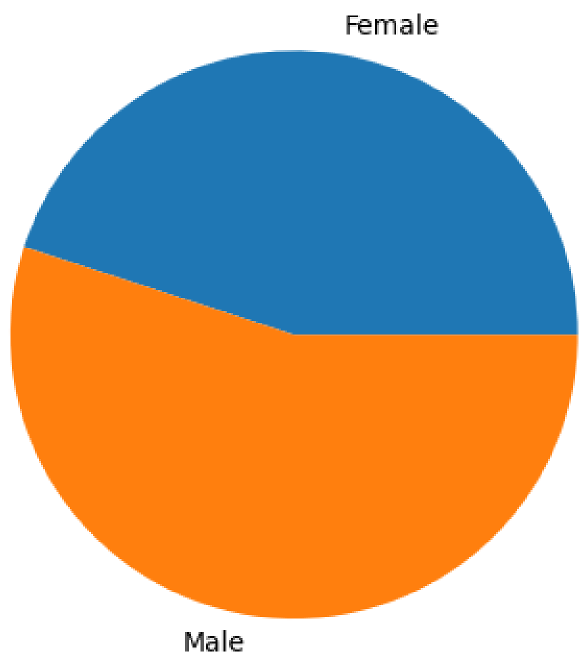
[74]:
```
sns.histplot(data=df, x='Age')
plt.show()
```



3.1.5 Question 3

3.1.6 Create a pie chart to show the proportion of 'Gender' in the dataset

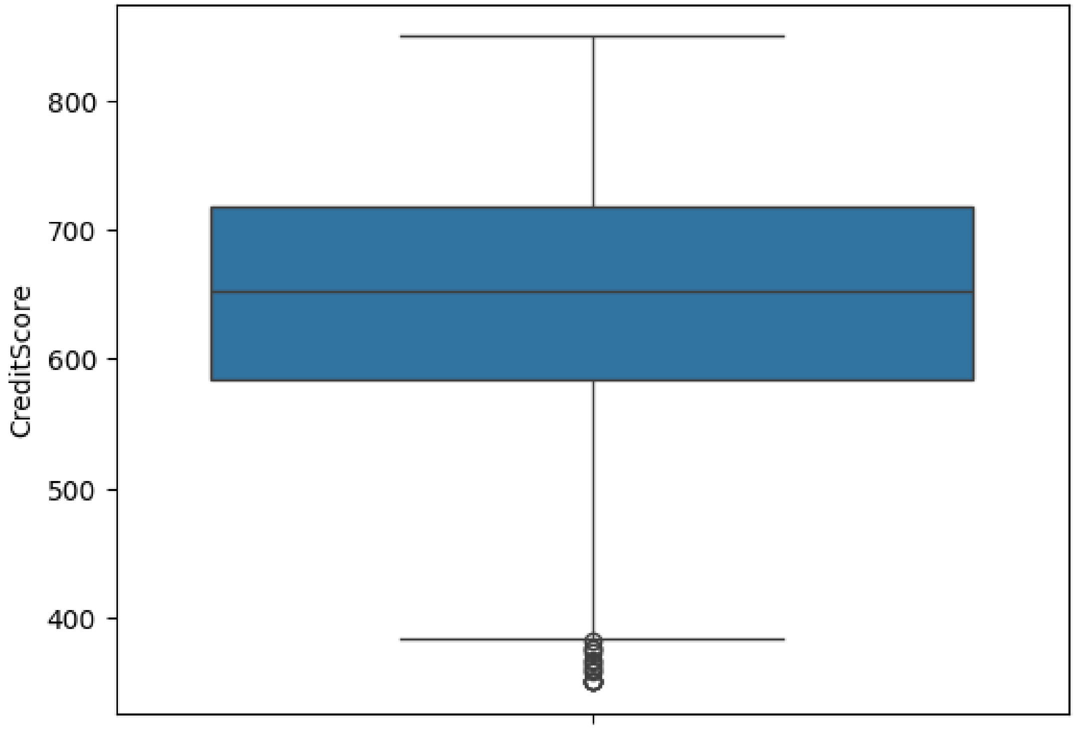
[75]:
```
# I know there must be a more efficient way to do this, but I missed the early␣
↪part of the lab.
pct_female = round(df[df['Gender'] == 'Female'].shape[0]/df.shape[0], 2)
pct_male = 1 - pct_female

plt.pie([pct_female, pct_male], labels=['Female', 'Male'])
plt.show()
```



3.1.7 Question 4

3.1.8 Create a boxplot for 'CreditScore' to identify outliers

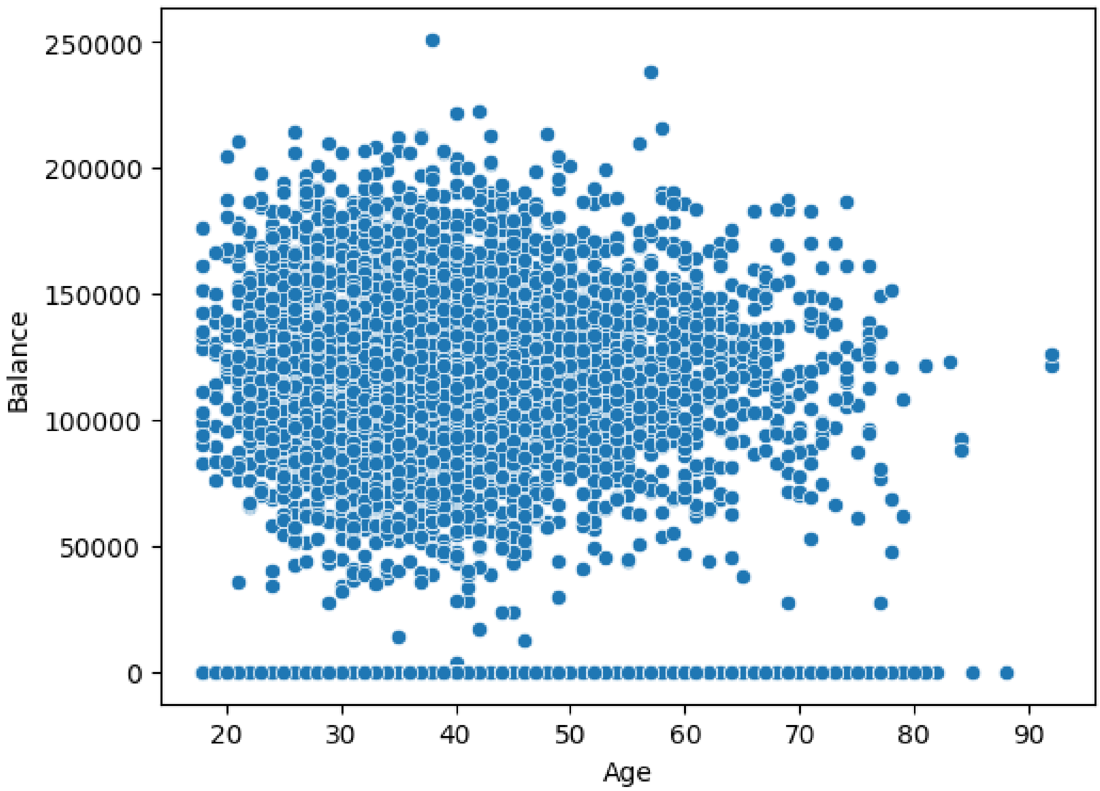
[76]:
```
# Seems odd to have a box plot for a single category numerical value like this
sns.boxplot(df, y='CreditScore')
plt.show()
```



3.1.9 Question 5

3.1.10 Create a scatter plot of 'Age' vs. 'Balance' to explore the relationship between these two variables

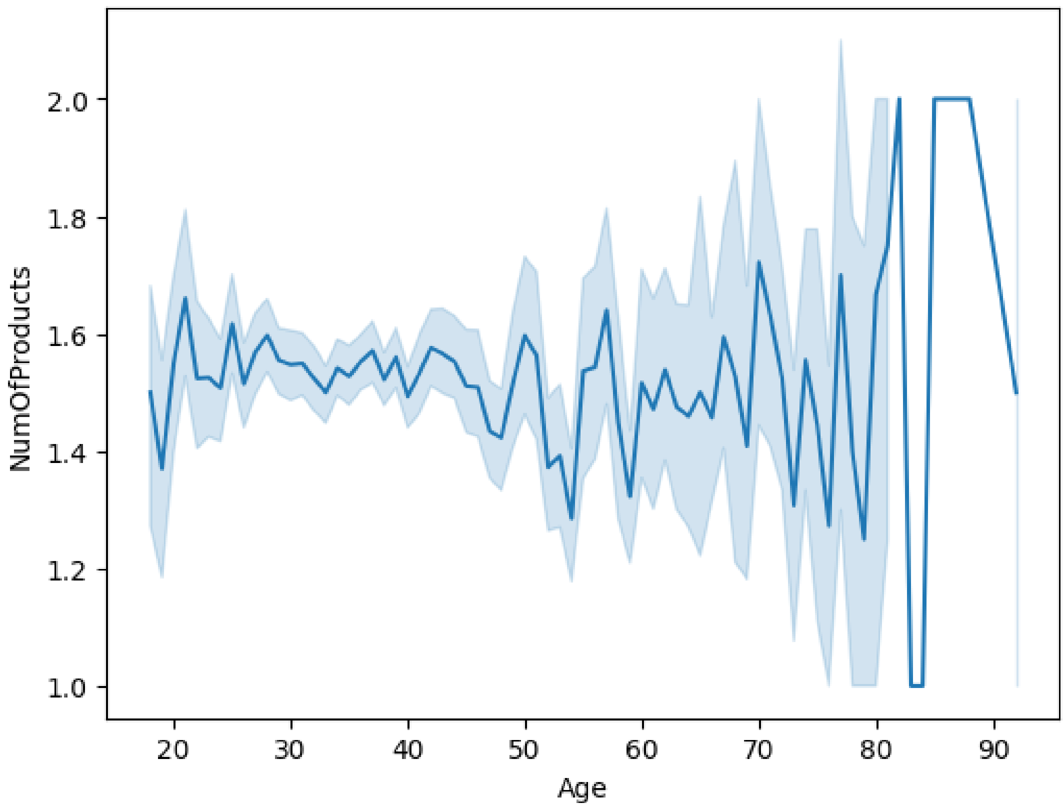
[77]:
```
sns.scatterplot(data=df, x='Age', y='Balance')
plt.show()
```



3.1.11 Question 6

3.1.12 Create a line chart to visualize the change in 'NumOfProducts' over 'Age'.

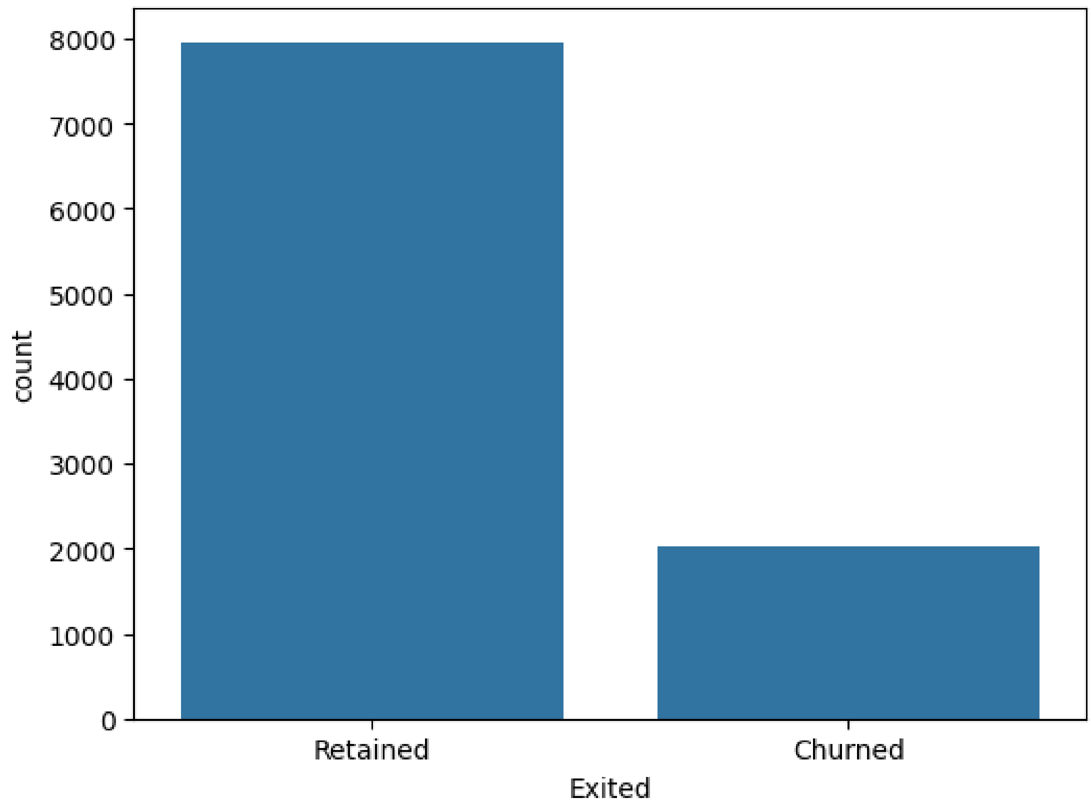
[78]:
```
sns.lineplot(df, x='Age', y='NumOfProducts')
plt.show()
```



3.1.13 Question 7

3.1.14 Create a bar chart to show the count of 'Exited' customers (churned vs. retained).

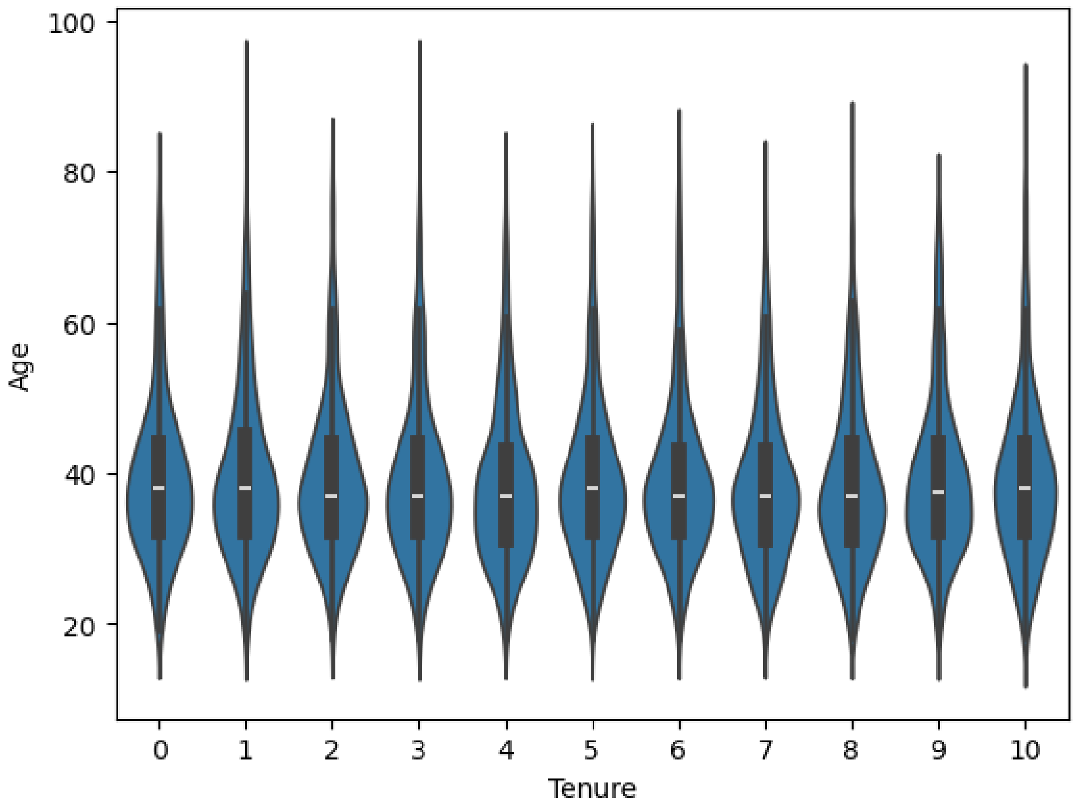
[79]:
```
# Took some research to get the x labels here but
# I'm not 100% sure what value means retained or churned.
ax = sns.countplot(data=df, x='Exited')
ax.set_xticks([0,1])
ax.set_xticklabels(['Retained', 'Churned'])
plt.show()
```



3.1.15 Question 8

3.1.16 Create a violin plot for 'Tenure' and 'Age' to compare their distributions.

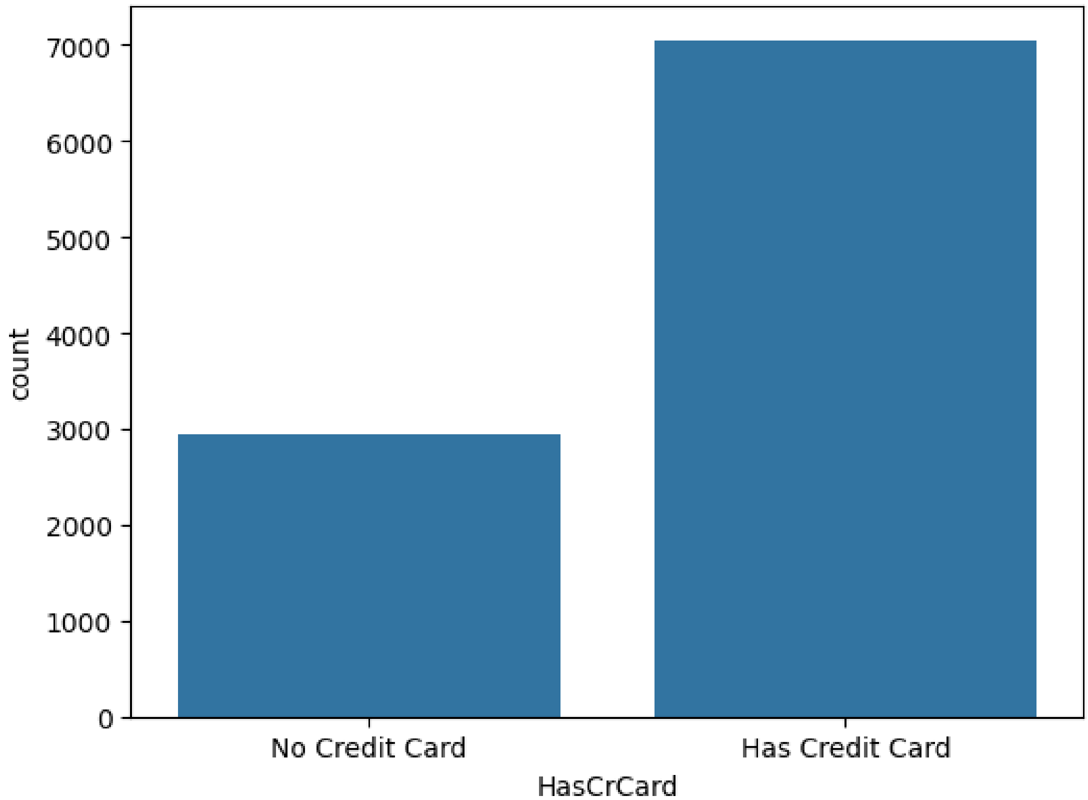
[80]:
```
sns.violinplot(data=df, x='Tenure', y='Age')
plt.show()
```



3.1.17 Question 9

3.1.18 Create a bar chart to show the count of 'HasCrCard' (having vs. not having a credit card).

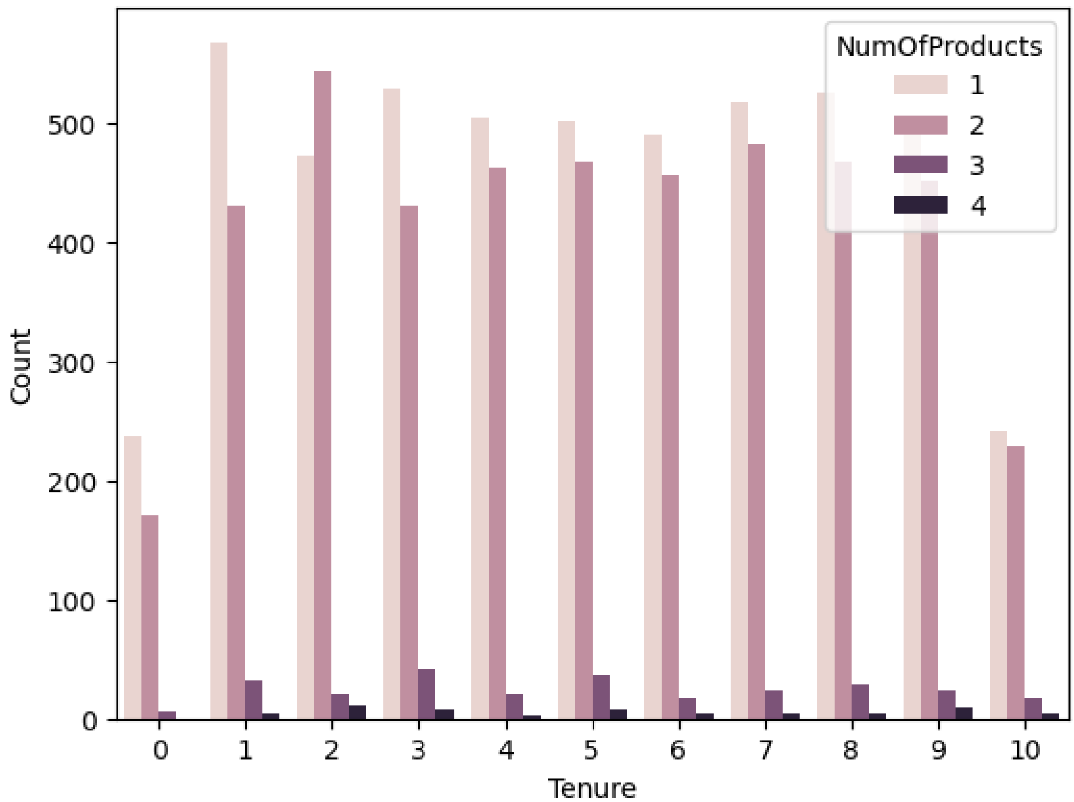
[81]:
```
ax = sns.countplot(data=df, x='HasCrCard')
ax.set_xticks([0, 1])
ax.set_xticklabels(['No Credit Card', 'Has Credit Card'])
plt.show()
```



3.1.19 Question 10

3.1.20 Create a bar chart to visualize the distribution of 'NumOfProducts' for customers with different 'Tenure' values.

[82]:
```
count_df = df.groupby(['Tenure', 'NumOfProducts']).size().
↪reset_index(name='Count') # This took a long time.

sns.barplot(data=count_df, x='Tenure', y='Count', hue='NumOfProducts')
plt.show()
```

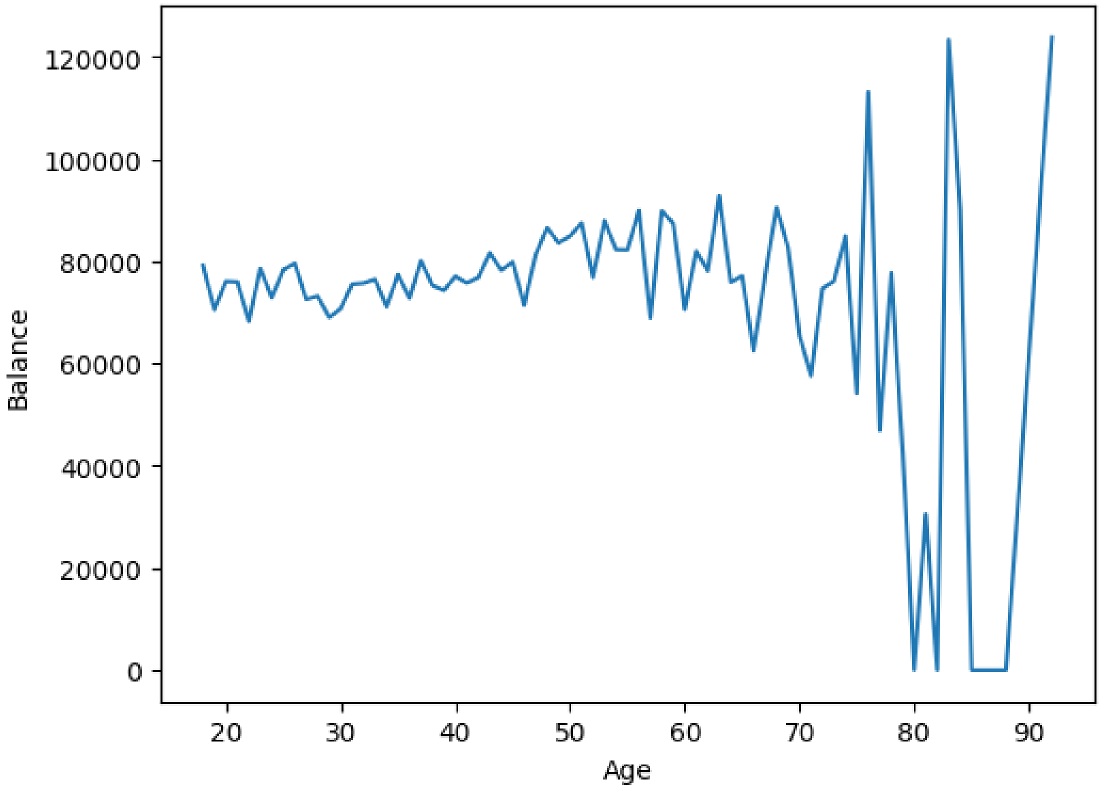


3.1.21 Question 11

3.1.22 How does the average balance of customers change with age? Create a line plot showing the average 'Balance' for each 'Age' group.

[83]:
```
# Converted to dataframe to suppress a warning about how data cannot be a␣
↪series.
balance_by_age = pd.DataFrame(df.groupby('Age')['Balance'].mean())

sns.lineplot(data=balance_by_age, x='Age', y='Balance')
```

[83]: <Axes: xlabel='Age', ylabel='Balance'>



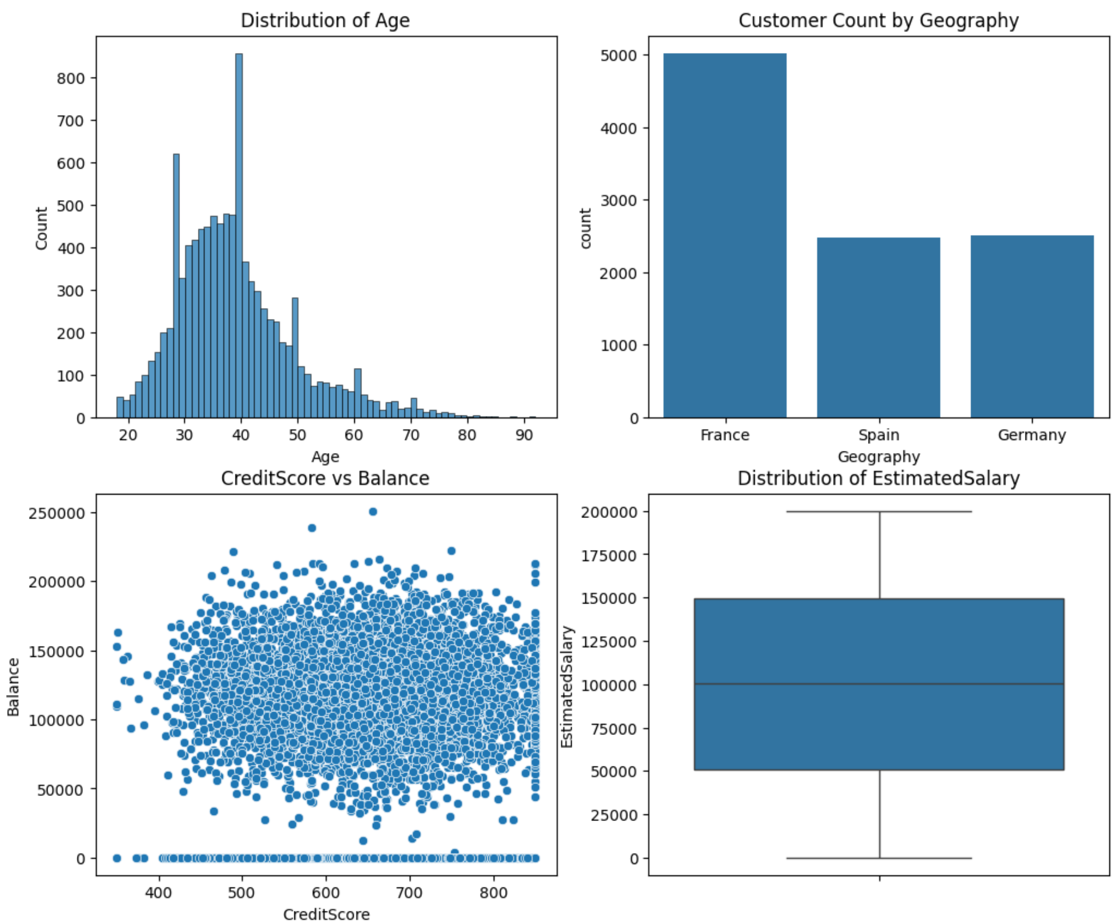
3.1.23 Question 12

3.1.24 Using the 'Churn' dataset, create a 2x2 grid of subplots. Each subplot should display one of the following:

- 1. A histogram showing the distribution of 'Age'.
- 2. A bar plot showing the count of customers in each 'Geography' category.
- 3. A scatter plot comparing 'CreditScore' and 'Balance'.
- 4. A boxplot showing the distribution of 'EstimatedSalary'.

[84]:
```
fig, axes = plt.subplots(2, 2, figsize=(12, 10))

# Histogram of Age
sns.histplot(data=df, x='Age', ax=axes[0, 0])
axes[0, 0].set_title('Distribution of Age')

# Bar plot of Geography count
sns.countplot(data=df, x='Geography', ax=axes[0, 1])
```

```
axes[0, 1].set_title('Customer Count by Geography')

# Scatter plot of CreditScore vs Balance
sns.scatterplot(data=df, x='CreditScore', y='Balance', ax=axes[1, 0])
axes[1, 0].set_title('CreditScore vs Balance')

# Boxplot of EstimatedSalary
sns.boxplot(data=df, y='EstimatedSalary', ax=axes[1, 1])
axes[1, 1].set_title('Distribution of EstimatedSalary')


plt.show()
```




```
[ ]:
```